The Square Case:
Gram Reduction and Spectral Transport for $N = d + 1$ Bodies

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Abstract

For N point masses in $\mathbb{R}^d$ the mass-weighted Jacobi configuration matrix $X = [\rho_1, \ldots, \rho_{N-1}]$ is $d \times (N-1)$. It is square exactly when $N = d+1$, and we show that this single coincidence organises the entire reduction. In the square case the Gram matrix $G = X^\top X$ is an unconstrained positive-semidefinite form, the trace-one slice $\Sigma_d = \{G \succeq 0,\ \mathrm{tr}G = 1\}$ is a smooth $\big(\frac{d(d+1)}{2} - 1\big)$-dimensional convex body, the rotation-reduced shape space is its orientation double cover branched over $\det X = 0$, and the tangent space splits canonically into $d-1$ eigenvalue directions and $\frac{d(d-1)}{2}$ eigenframe directions. At $d = 2$ this reproduces the Kendall shape sphere with its standard radius $1/2$, equator the collinear locus, in three lines; at $d = 3$ it gives the five-dimensional four-body shape space with the coplanar locus as branch divisor.

On this ladder we study the intrinsic transport operator $\mathcal{A}(H) = TH + HT - 2\mathrm{tr}(TH)\, G$ acting on symmetric traceless H. It is block diagonal in the eigenvalue/eigenframe split; its eigenframe spectrum is exactly $\{t_i + t_j\}$; and its eigenvalue block is a rank-one modification of a diagonal matrix whose characteristic equation is the secular equation $\sum_i g_i t_i/(t_i - \lambda) = 1$. Because the weights $g_i t_i$ are positive, that equation has one root strictly between each pair of consecutive t_i. Hence for every d, and on the whole regular stratum, $\mathcal{A}$ is non-normal with strictly real, interlacing spectrum: exceptional points cannot occur in the interior. They occur on the boundary $\det G = 0$, where an eigenvalue is pinned at $2t_k$ and a secular root may meet it, producing a genuine Jordan block. Binary collision loci are shown to have codimension d, not one.

Two structural facts sharpen this. First, the operator is defined by the matrix trace alone and uses no inner product: its spectrum is unchanged in a non-orthonormal basis, so the results are metric-free. Second, the non-symmetric rank-one update is diagonally similar to a symmetric one, $D_t - gt^\top = S(D_t - uu^\top)S^{-1}$ with $u = \sqrt{g \odot t}$ and $S = \mathrm{diag}\sqrt{g/t}$, which gives reality and diagonalisability by a second route and identifies the inner product in which the operator is self-adjoint. That similarity degenerates exactly on the boundary, which is why defectiveness can appear only there.

A final part takes the metric out of the primitives altogether. On the cone that Σ_d slices, the Koszul–Vinberg characteristic function requires only a volume form, and for positive definite symmetric matrices its logarithm is $-(d+1)\log \mathrm{Vol}$ with $\mathrm{Vol} = |\det X|$ the simplex volume: the metric is the Hessian of the log volume rather than assumed data. In that structure the free tensor is no longer free. Both operator families are self-adjoint for the induced metric exactly when their parameter is the dual coordinate G^{-1}, which settles the one thing left open above. The boundary behaviour then changes completely — eigenvalues diverge as inverse powers of the volume rather than pinning at finite values, at two distinct rates Vol^{-2} and Vol^{-1}, with no defectiveness — so that analysis is replaced rather than extended, and we say so. Three further steps close the volume part. The congruence operator is the unique positive square root of the volume metric itself, read as an operator through the trace pairing — a kinematic selection between the two canonical operators, with the additive one identified as its logarithmic generator. The cubic form is the connection: in the flat coordinates the Christoffel symbols of

the volume metric are exactly half the third derivative of the potential, and the geodesics are $G^{1/2}e^{tV}G^{1/2}$ in closed form. And at the corner of the divisor, where two eigenvalues vanish together, the two divergence rates collide into one: the divisor is smooth precisely where the rates split and singular precisely where they merge. Off the ladder, the tangent split survives in threefold form — eigenvalue, frame-within-support, and support-moving blocks — and the last block, which has no interior analogue, is the precise boundary of what is established here.

We also mark where the construction stops. For $N-1>d$ the Gram matrix is confined to a determinantal variety of rank at most d, the tangent split fails, and the results here do not extend; the first such case, five bodies in three dimensions, is described explicitly. All statements are verified numerically and the verification script is included.

1 Introduction

Reductions of the N-body problem by translation, rotation and scale are classical, and the two best known low cases — three bodies in the plane and four bodies in space — are usually treated as separate constructions of increasing difficulty. They are not. They are the same construction at two values of one parameter, and the parameter is not N but the spatial dimension d.

The observation is elementary. After removing the centre of mass, a configuration of N bodies in $\mathbb{R}^d$ is a $d\times(N-1)$ matrix X of mass-weighted Jacobi vectors. This matrix is square precisely when

$$N = d+1,$$

that is, when the bodies form a full d-simplex. Squareness is not cosmetic: it is the exact condition under which the rotation-invariant Gram matrix $G = X^\top X$ is an *unconstrained* symmetric positive-semidefinite form, under which the determinant $\det X$ exists and supplies an orientation, and under which the chain

$$Nd \longrightarrow (N-1)d \longrightarrow \tfrac{d(d+1)}{2} \longrightarrow \tfrac{d(d+1)}{2} - 1$$

closes with no remainder. Off this diagonal the Gram matrix is pinned to rank at most d, and every statement below fails.

d	N	X	after c.o.m.	after rotations	shape dimension
2	3	2×2	4	3	**2** (Kendall shape sphere)
3	4	3×3	9	6	**5**
4	5	4×4	16	10	**9**
5	6	5×5	25	15	**14**
3	5	3×4	12	9	8 (off ladder; G singular)

Table 1: The square ladder $N = d+1$ and the first case off it.

What this paper does. Sections 2–3 set up the Gram reduction uniformly in d, identify the rotation-reduced shape space as a branched double cover, and locate the strata — including the fact, easy to miss, that binary collision loci have codimension d rather than one. Section 4 isolates the canonical splitting of the tangent space into eigenvalue and eigenframe directions and identifies the eigenframe rotation rate with the first-order spectral response of a symmetric matrix. Sections 5–6 define the transport operator and prove its spectral structure, which is the main result. Section 8 states where the ladder ends and why.

The volume part. Section 11 closes what the metric sections leave open, at the kinematic level: the parameter of the operator family is forced to the dual coordinate G^{-1} (Theorem 11.2); the congruence operator is the unique positive square root of the volume-metric operator and the additive one its logarithmic generator (Theorem 11.6); the cubic form is exactly twice the Christoffel data of the volume metric, with geodesics in closed form (Theorem 11.8); the boundary rates split on the smooth divisor and merge at its corner (Theorem 11.10, Proposition 11.13); and off the ladder the tangent split survives threefold (Proposition 8.1).

What is assumed and what is not. The operator below depends on one symmetric positive tensor T commuting with G. Every theorem is proved for arbitrary such T; we do not require a particular choice. Section 7 discusses the canonical closure $T = G$, which is the only choice constructible from the shape alone, and states plainly that any other T requires its own derivation, which this paper does not supply.

2 The square case and the Gram reduction

2.1 Setup

Let $q_1, \dots, q_N \in \mathbb{R}^d$ with masses $m_1, \dots, m_N$, centre of mass removed, and let $\rho_1, \dots, \rho_{N-1}$ be the mass-weighted Jacobi vectors, normalised so that the kinetic energy is $\frac{1}{2}\sum_k |\dot{\rho}_k|^2$ and the moment of inertia is $\sum_k |\rho_k|^2$. Assemble

$$X = [\rho_1, \dots, \rho_{N-1}] \in \mathbb{R}^{d\times(N-1)}, \qquad N = d+1 \;\Rightarrow\; X \in \mathbb{R}^{d\times d}.$$

A rotation of physical space acts by $X \mapsto RX$, $R \in SO(d)$; a scaling by $X \mapsto \mu X$.

Proposition 2.1 (The invariant)**.** *$G := X^\top X$ is invariant under $X \mapsto RX$ for every $R \in O(d)$, and G determines X up to that action. The equivariant matrix $XX^\top$ is* not *invariant: it transforms as $RXX^\top R^\top$.*

Proof. $(RX)^\top(RX) = X^\top R^\top RX = X^\top X$. Conversely if $Y^\top Y = X^\top X$ with X invertible then $Q := YX^{-1}$ satisfies $Q^\top Q = I$. □

The two matrices share a spectrum, which is why an argument conducted entirely in a principal frame will not detect a confusion between them; the distinction is nevertheless decisive, since only one of them is a function on shape space.

Definition 2.2. $\Sigma_d := \{G \in \mathrm{Sym}(d) : G \succeq 0,\ \mathrm{tr} G = 1\}$.

Proposition 2.3 (Dimension)**.** *$\dim \Sigma_d = \frac{d(d+1)}{2} - 1$, and this equals $(N-1)d - \frac{d(d-1)}{2} - 1$, the count obtained by removing rotations and scale.*

Consequently $SO(d)$ acts *trivially* on Σ_d: the group has already been quotiented out. The principal bundle in play is

$$P = \{X \in \mathbb{R}^{d\times d} : \det X \neq 0\} \;\longrightarrow\; \Sigma_d,$$

with $SO(d)$ acting on the total space. There is no horizontal restriction on $T_G\Sigma_d$; every tangent direction of Σ_d is realised by a motion of the bodies. The connection on P governs reconstruction of the physical rotation, not admissibility of shape velocities.

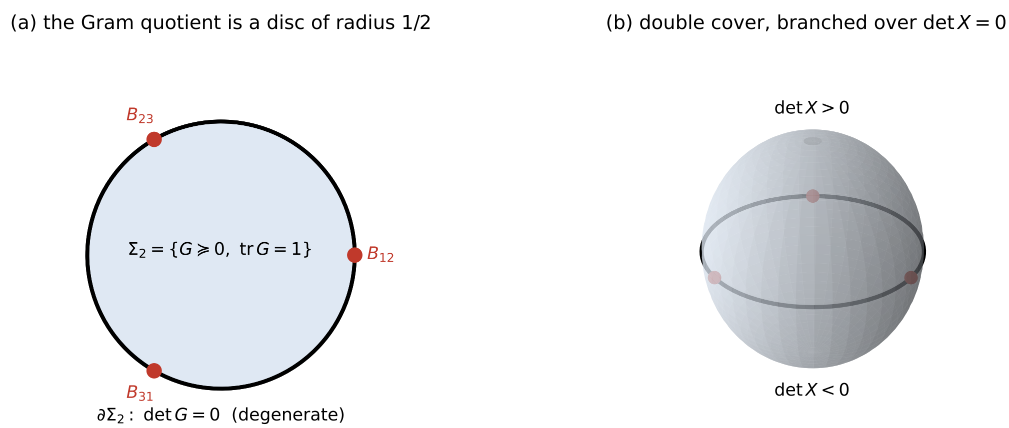

Figure 1: The construction at $d = 2$. (a) Σ_2 is a disc of radius $1/2$; its boundary is $\det G = 0$, the degenerate (collinear) configurations, and the three binary collision loci are the marked boundary points. (b) The rotation-reduced shape space is the double cover branched over that boundary, i.e. the sphere, with the two hemispheres carrying the two signs of $\det X$.

2.2 Orientation and the branched double cover

Theorem 2.4 (Branched cover). *Σ_d is the quotient of P (normalised) by $O(d)$. The rotation-reduced shape space, the quotient by $SO(d)$, is the double cover of Σ_d branched over the divisor*

$$\mathcal{D} := \{\det G = 0\} = \{\det X = 0\},$$

the configurations that fail to span $\mathbb{R}^d$.

Proof. By Proposition 2.1, $Y^\top Y = X^\top X$ implies $Y = QX$ with $Q \in O(d)$, and $\det Q = \det Y / \det X$. If $\det X \neq 0$ the two $SO(d)$-orbits with the same G are distinguished by the sign of $\det X$; if $\det X = 0$ they coincide. □

2.3 $d = 2$: the shape sphere, with its radius

Write $G = \begin{pmatrix} \frac{1}{2} + u & v \\ v & \frac{1}{2} - u \end{pmatrix}$, which is the general trace-one symmetric 2×2 matrix. Then

$$\det G = \tfrac{1}{4} - u^2 - v^2,$$

so Σ_2 is exactly the disc $u^2 + v^2 \leq \frac{1}{4}$. Adjoining the orientation coordinate $w_3 := \det X$, and using $\det G = (\det X)^2$,

$$u^2 + v^2 + w_3^2 = \tfrac{1}{4}. \tag{1}$$

Corollary 2.5. *For three bodies in the plane the rotation-reduced, scale-fixed shape space is the sphere of radius $\frac{1}{2}$, with $w_3 = \det X$ the signed area, the equator $w_3 = 0$ the collinear locus, and the two hemispheres the two orientations.*

This is the Kendall shape sphere in the normalisation in which the reduced kinetic metric is $\frac{1}{4}(\mathrm{d}w_1^2 + \mathrm{d}w_2^2 + \mathrm{d}w_3^2)$, recovered here without choosing a Hopf map. The three binary collision loci sit on the equator, alternating with the three Euler configurations.

2.4 Two metrics, and which one is which

Σ_d carries two natural metrics and they are not the same, so the choice must be stated once and kept.

- The **Frobenius** structure $\langle H, K\rangle = \mathrm{tr}(HK)$ on $\mathrm{Sym}(d)$. This is what makes the basis of Section 4 orthonormal and what the transport operator below is written in. At $d = 2$ it is flat: $\|\delta G\|_F^2 = 2(\mathrm{d}u^2 + \mathrm{d}v^2)$ on the disc.
- The **kinetic** (submersion) metric induced from the flat metric on X, which is the physically reduced kinetic energy. At $d = 2$, pushing an orthonormal horizontal frame forward to the coordinates (1) gives the Gram matrix $\mathrm{diag}(1, 1)$ with off-diagonal 7×10^{-12}: the kinetic metric is exactly the round metric of the radius-$\frac{1}{2}$ sphere.

The interior of Σ_d is also diffeomorphic to $SL(d, \mathbb{R})/SO(d)$, whose standard affine-invariant metric is neither of the above — at $d = 2$ it is the hyperbolic plane. The identification is useful for the manifold; the metric that comes with it is not the one used here, and importing it silently would change every geodesic in the theory. (Section 11 returns to this point: the volume structure turns out to *derive* exactly this metric rather than import it, which changes the status of the warning — see the remark after Theorem 11.8.)

3 Stratification

3.1 Rank strata

Σ_d is stratified by $r = \mathrm{rank}\, G = \mathrm{rank}\, X$, the dimension actually spanned by the configuration. The stratum $\{r \le k\}$ is a copy of $\{$trace-one positive semidefinite forms on a k-dimensional space$\}$ bundled over the Grassmannian of k-planes; its dimension is $\frac{k(k+1)}{2} - 1 + k(d - k)$, and the top boundary $r = d - 1$ has codimension one.

d	$r = d - 1$ (codim 1)	physical meaning
2	rank 1	collinear
3	rank 2	coplanar
4	rank 3	confined to a hyperplane

Table 2: The codimension-one boundary divisor $\mathcal{D}$ at low d. At $d = 3$ it is the *coplanar* locus; the collinear configurations are rank 1, of codimension three.

The rotation action on P is free where $r \ge d-1$ and acquires isotropy $SO(d-r)$ where $r \le d-2$. At $d = 2$ there is no such stratum inside Σ_2 other than the rank-one boundary; at $d = 3$ the collinear locus $r = 1$ carries $SO(2)$ isotropy and is the genuine singular locus of the reduction.

3.2 Collision loci have codimension d

Let $q_i - q_j = X c^{(ij)}$ for a fixed vector $c^{(ij)} \in \mathbb{R}^d$ determined by the masses. Then

$$\sigma_{ij}(G) := \left(c^{(ij)}\right)^{\top} G\, c^{(ij)} = |q_i - q_j|^2. \tag{2}$$

Proposition 3.1. *$\sigma_{ij}(G) = 0$ if and only if $c^{(ij)} \in \ker G$. Hence the binary collision locus $\mathcal{B}_{ij}$ lies inside $\mathcal{D}$ and*

$$\dim \mathcal{B}_{ij} = \tfrac{d(d-1)}{2} - 1, \qquad \mathrm{codim}\, \mathcal{B}_{ij} = d.$$

Proof. $G \succeq 0$, so $c^\top Gc = 0$ iff $Gc = 0$. The trace-one positive semidefinite forms annihilating a fixed vector are the trace-one forms on $\mathbb{R}^d/\langle c\rangle$. $\square$

At $d = 2$ this gives $\dim \mathcal{B}_{ij} = 0$: the binary collisions are the three marked *points* on the equator, as they should be. At $d = 3$ it gives $\dim \mathcal{B}_{ij} = 2$ in a five-dimensional space. In particular collisions are never hypersurfaces, and a generic trajectory does not meet them — the classical fact, recovered here from the rank condition.

The vectors $c^{(ij)}$ at $d = 3$ are recorded in Appendix A; all six are verified against (2) directly.

4 The canonical tangent splitting

$T_G\Sigma_d$ is the space of symmetric traceless $d \times d$ matrices. In a principal frame $G = \mathrm{diag}(g_1, \ldots, g_d)$ it splits orthogonally under the Frobenius product:

$$\underbrace{\tfrac{d(d+1)}{2} - 1}_{\dim \Sigma_d} = \underbrace{(d-1)}_{\text{diagonal: change the } g_i} + \underbrace{\tfrac{d(d-1)}{2}}_{\text{off-diagonal: rotate the frame}}. \tag{3}$$

The identity holds for every d: $2 = 1 + 1$, $5 = 2 + 3$, $9 = 3 + 6$, $14 = 4 + 10$, $20 = 5 + 15$. The first summand deforms the shape of the inertia ellipsoid, the second rotates its principal axes.

Lemma 4.1 (The eigenframe rate is the spectral response)**.** *Let $G = \mathrm{diag}(g)$ with $g_i \neq g_j$ and let δG be a symmetric perturbation. To first order the i-th eigenvector rotates toward the j-th by the angle*

$$\Omega_{ij} = \frac{\delta G_{ij}}{g_i - g_j}.$$

This is first-order eigenvector perturbation theory, and it is exactly the spectral response formula $\langle m|V|n\rangle = \langle m|\delta\Sigma|n\rangle/(\lambda_{\mathrm{in}} - \lambda_{\mathrm{out}})$ of the projector framework, specialised to real symmetric matrices. It is worth saying plainly that Ω is *not* the mechanical connection of the bundle $P \to \Sigma_d$: it is a statement about eigenvectors of G, involves no group action, and lives entirely on the base. Measured against the true eigenvectors of $G + \delta G$ it reproduces the predicted rotation to ratio 1.000000 at $\varepsilon = 10^{-4}$ and 10^{-5}.

5 The transport operator

Definition 5.1 (Transport operator)**.** Let $T \in \mathrm{Sym}(d)$ be positive definite and commute with G. On $H \in T_G\Sigma_d$ set

$$\mathcal{A}(H) \ := \ TH + HT - 2\,\mathrm{tr}(TH)\,G. \tag{4}$$

Lemma 5.2 (The coefficient is forced)**.** $\mathrm{tr}\big(TH + HT - c\,\mathrm{tr}(TH)G\big) = (2 - c)\mathrm{tr}(TH)$*, so $\mathcal{A}$ maps $T_G\Sigma_d$ to itself if and only if $c = 2$.*

Numerically, with $c = 1$ the image leaves the tangent space by 0.693 in the test case; with $c = 2$ by 2.8×10^{-16}.

Proposition 5.3 (Block structure)**.** *In a principal frame of G, with T diagonal, $\mathcal{A}$ preserves the splitting* (3)*: it is block diagonal, with no coupling between eigenvalue and eigenframe directions.*

Proof. For off-diagonal H, $\mathrm{tr}(TH) = 0$, so $\mathcal{A}(H) = TH + HT$, again off-diagonal. For diagonal H all three terms are diagonal. $\square$

Proposition 5.4 (Rank-one form on the eigenvalue block)**.** *On diagonal $H = \operatorname{diag}(h)$ with $\sum_i h_i = 0$,*

$$\mathcal{A}(H) = 2\operatorname{diag}\big((D_t - g\,t^\top)h\big), \qquad D_t = \operatorname{diag}(t),$$

so the eigenvalue block is twice the restriction of a rank-one modification of a diagonal matrix.

5.1 The operator uses no metric

Proposition 5.5 (Metric-freeness)**.** *The map (4) is defined using only the matrix product and the matrix trace. No inner product on $\mathrm{Sym}_0(d)$ enters, and the requirement $\operatorname{tr}\mathcal{A}(H) = 0$ fixes the coefficient without reference to one. Consequently the spectrum of $\mathcal{A}$ is independent of the basis used to represent it.*

Numerically, representing $\mathcal{A}$ in a random non-orthonormal basis of $\mathrm{Sym}_0(d)$ reproduces the same spectrum to 2.1×10^{-9} across $d = 3, 4, 5$. The Frobenius basis of Section 4 is bookkeeping, not structure. This matters beyond tidiness: the metric ambiguity flagged in Section 2.4 does not propagate into any spectral statement below.

5.2 The operator is diagonally similar to a symmetric one

Theorem 5.6 (Symmetrisation)**.** *Let t, g have strictly positive entries and set $u := \sqrt{g \odot t}$ (entry-wise) and $S := \operatorname{diag}\big(\sqrt{g_i/t_i}\big)$. Then*

$$D_t - g\,t^\top \;=\; S\left(D_t - uu^\top\right)S^{-1}.$$

Hence the eigenvalue block of $\mathcal{A}$ is similar to a symmetric rank-one modification of a diagonal matrix. It therefore has real spectrum, is diagonalisable, and is self-adjoint for the inner product $\langle h, k\rangle_S = \sum_i h_i k_i\, t_i/g_i$.

Proof. $(Su)_i = \sqrt{g_i/t_i}\,\sqrt{g_i t_i} = g_i$ and $(u^\top S^{-1})_i = \sqrt{g_i t_i}\,\sqrt{t_i/g_i} = t_i$, so $S(uu^\top)S^{-1} = g\,t^\top$, while S commutes with D_t. □

Verified to 1.3×10^{-15} over 10^4 random draws at $d = 2, \ldots, 6$.

Remark 5.7 (Where the hidden inner product fails). S is defined only where every $g_i > 0$, that is, on the interior. On the divisor $\mathcal{D}$ some g_k vanishes, S degenerates, and self-adjointness is lost. This is the mechanism behind Theorem 6.5: the operator carries its own inner product on the interior, and the boundary is exactly where that inner product ceases to exist. Nothing external is being imposed; the failure is intrinsic.

6 The spectrum

Theorem 6.1 (Eigenframe spectrum)**.** *The eigenframe block of $\mathcal{A}$ is diagonal with eigenvalues $t_i + t_j$, $i < j$. In particular it carries no dependence on G at all.*

Proof. $TE_{ij} + E_{ij}T = (t_i + t_j)E_{ij}$ for E_{ij} the symmetrised elementary matrix. □

Theorem 6.2 (Secular equation, interlacing, reality)**.** *Let $T \succ 0$ and $G \succ 0$ with $\operatorname{tr}G = 1$. The eigenvalue block of $\mathcal{A}$ has spectrum $\{2\lambda_1, \ldots, 2\lambda_{d-1}\}$ where the λ_k are the roots, other than $\lambda = 0$, of*

$$\sum_{i=1}^{d} \frac{g_i t_i}{t_i - \lambda} = 1. \tag{5}$$

Ordering $t_{(1)} < \cdots < t_{(d)}$, there is exactly one root in each interval $(t_{(k)}, t_{(k+1)})$, so

$$t_{(1)} < \lambda_1 < t_{(2)} < \lambda_2 < \cdots < \lambda_{d-1} < t_{(d)}.$$

Consequently the spectrum of $\mathcal{A}$ is real on the whole regular stratum, and $\mathcal{A}$ has no exceptional points there.

Proof. By Proposition 5.4 the eigenvalue block is $2(D_t - gt^\top)$ restricted to $\{\sum h_i = 0\}$, which is invariant. The characteristic polynomial of the unrestricted matrix is

$$\det\left(D_t - gt^\top - \lambda I\right) = \prod_i (t_i - \lambda) \cdot \left(1 - \sum_i \frac{g_i t_i}{t_i - \lambda}\right),$$

and $\lambda = 0$ is always a root since $\sum_i g_i = 1$; it corresponds to the trace direction, which $\mathcal{A}$ annihilates. Put $f(\lambda) = \sum_i g_i t_i/(t_i - \lambda)$. Then

$$f'(\lambda) = \sum_i \frac{g_i t_i}{(t_i - \lambda)^2} > 0$$

because every weight $g_i t_i$ is strictly positive. Hence f is strictly increasing on each interval between consecutive poles, running from $-\infty$ to $+\infty$, and meets the level 1 exactly once there. That accounts for $d-1$ real roots, and with $\lambda = 0$ for all d. Reality of the eigenframe block is Theorem 6.1. Since all $d-1$ eigenvalue-block roots are simple by strict interlacing, no Jordan block can form. □

Remark 6.3 (What is classical here and what is not). The secular equation for a rank-one modification of a diagonal matrix, and the strict interlacing that follows from monotonicity between poles, are classical: they are the basis of the divide-and-conquer eigensolver and go back to Golub [9] and Bunch, Nielsen and Sorensen [10]. We claim none of that. What is claimed here is the identification — that the transport operator of the square ladder *is* such an update, with weights $g_i t_i$ that are positive for the structural reason that G and T are positive — together with Theorem 5.6, which brings the non-symmetric case in this problem under the classical symmetric theorem by an explicit diagonal similarity. The classical hypothesis we are outside of is symmetry of the update; the classical hypothesis we keep, and which does all the work, is positivity of the weights.

Remark 6.4 (Non-normal, but real). $\mathcal{A}$ is genuinely non-normal: the eigenvalue block is $2(D_t - gt^\top)$, which is not symmetric unless g and t are proportional. The measured commutator norm $\|[\mathcal{A}, \mathcal{A}^\top]\|$ vanishes when T is isotropic *and also* when G is isotropic, and is nonzero only when both are anisotropic. Non-normality is therefore a joint property of the mass-inertia tensor and the shape, not of T alone. Theorem 6.2 says that this non-normality never produces complex eigenvalues in the interior — the two properties are independent here, which is not automatic for a non-normal family.

6.1 The boundary: pinning and exceptional points

Theorem 6.5 (Boundary spectrum). *Let $Z = \{k : g_k = 0\}$ be non-empty, so G lies on the divisor $\mathcal{D}$. Writing $M = D_t - g\,t^\top$ and $\mathcal{A}v = 2Mv$ on the traceless subspace:*

(i) For each $k \in Z$, $\lambda = t_k$ is an eigenvalue of M with a genuine eigenvector; the corresponding eigenvalue of $\mathcal{A}$ is pinned *at $2t_k$.*

(ii) The remaining eigenvalues are the roots of the reduced secular equation $\sum_{i \notin Z} g_i t_i/(t_i - \lambda) = 1$.

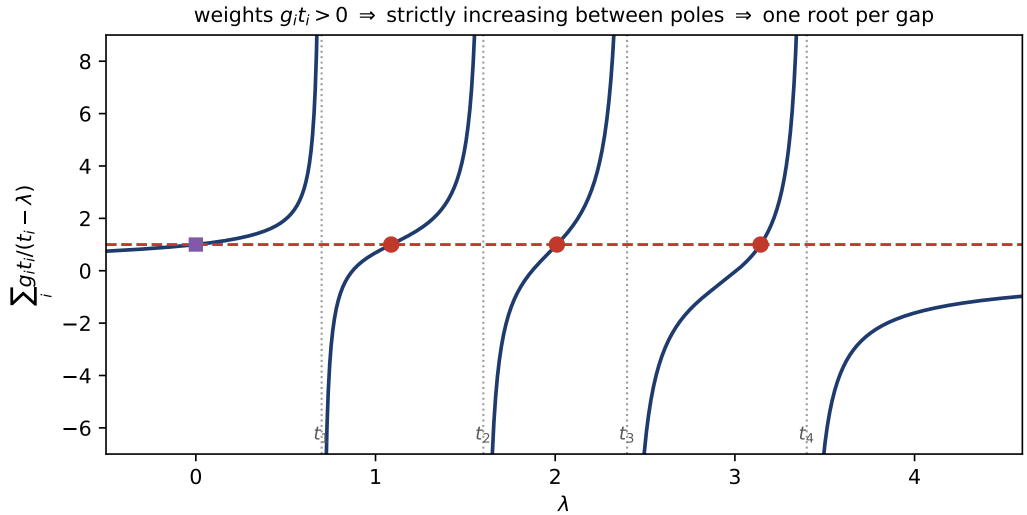

Figure 2: Theorem 6.2 in one picture, at $d = 4$. The weights $g_i t_i$ are positive, so the secular function increases strictly between poles; it therefore crosses the level 1 (dashed) exactly once in each gap between consecutive t_i. The roots (circles) strictly interlace the poles; the square marks the root at $\lambda = 0$ belonging to the trace direction.

(iii) A Jordan block forms precisely when a root of (ii) coincides with a pin t_k. Coincidence of two pins, $t_k = t_l$ with $k, l \in Z$, is semisimple*: the eigenspace is two-dimensional and no defect occurs.*

Proof. $Mv = D_t v - (t \cdot v)\, g$. If $t \cdot v \neq 0$, normalise it to 1; then $v_i = g_i/(t_i - \lambda)$ and the constraint reproduces the secular equation, giving (ii). For $k \in Z$ take $\lambda = t_k$: the components v_i, $i \neq k$, are determined as above and v_k remains free, so $t \cdot v = 1$ can be solved for v_k since $t_k > 0$; this gives (i). If $t \cdot v = 0$ then $t_i v_i = \lambda v_i$, so v is supported on $\{i : t_i = \lambda\}$, which for two coinciding pins is two-dimensional, giving the semisimple case in (iii). The defective case is the branch collision between (i) and (ii). □

Verified at $d = 3$: with $t = (1.9, 0.5, 1.2)$ and $g = (0.6, 0.4, 0)$ the eigenvalue block has spectrum $\{2.12,\ 2.40\}$ with $2t_3 = 2.40$ pinned as predicted; tuning t_3 so that the secular root meets the pin gives a repeated eigenvalue with normalised eigenvector determinant 6.1×10^{-16} — defective. By contrast, at $g = (1, 0, 0)$ and $t_2 = t_3$, where two pins coincide, the same measurement returns 1.000: semisimple, exactly as (iii) requires. Pins at one, two and three vanishing eigenvalues were checked at $d = 3, 4, 5$ and appear in the spectrum in every case.

Remark 6.6 (Genericity). The defective locus is cut out by two conditions — $\det G = 0$ and the branch collision — so it has codimension two in the combined (G, T) data. A generic one-parameter path therefore misses it, while a generic two-parameter family meets it at isolated points. Perturbing G off the divisor removes the defect immediately: at perturbation amplitudes 0, 10^{-6}, 10^{-4}, 10^{-2} the normalised eigenvector determinant moves $6 \times 10^{-16} \to 2.8 \times 10^{-3} \to 4.5 \times 10^{-2} \to 3.4 \times 10^{-1}$.

The picture is therefore the opposite of what one might guess: the interior is spectrally benign for every d, and all defectiveness is pushed onto the codimension-one divisor $\mathcal{D}$ and cut down further there. By the remark after Theorem 5.6, the reason is that the operator supplies its own inner product on the interior and loses it exactly on $\mathcal{D}$.

7 What T is

Everything above holds for any positive definite T commuting with G. The theory is closed — that is, requires no data beyond the shape — for exactly one choice:

$$T = G, \qquad \mathcal{A}(H) = GH + HG - 2\mathrm{tr}(GH)\,G.$$

Then $t_i = g_i$, the eigenframe spectrum is $\{g_i + g_j\}$, and Theorem 6.2 applies verbatim with the weights $g_i^2 > 0$. This choice is superseded in Section 11, where the volume structure forces $T \propto G^{-1}$ instead, and forces it rather than offering it. There is a cost to $T = G$ that must be stated: at $T = G$ the similarity of Theorem 5.6 is the identity, and $\mathcal{A}$ becomes *symmetric*. Measured over $d = 2, \dots, 5$, $\|\mathcal{A} - \mathcal{A}^\top\|$ is 3×10^{-16} at $T = G$ and of order unity otherwise. So the canonical closure and non-normality are mutually exclusive: any programme that wants a non-normal transport operator on this ladder needs $T \neq G$, and therefore needs a derivation of T. We record this as the canonical closure and we do not claim it is forced by mechanics. Any other T — in particular any T carrying velocity or driving data — must be derived from a stated dynamical principle. No such derivation exists in this framework at present, and the results above should be read as a statement about the family of operators (4), parametrised by T, rather than about one distinguished dynamical operator.

8 Where the ladder ends

For $N - 1 > d$ the configuration matrix X is $d \times (N-1)$ with $N - 1 > d$, and $G = X^\top X$ is an $(N-1) \times (N-1)$ Gram matrix of rank at most d. Three things break simultaneously.

1. G **is constrained.** It no longer ranges over an open subset of $\mathrm{Sym}(N-1)$ but over the determinantal variety $\{\mathrm{rank} \le d\}$, which is singular along its own deeper strata. At $d = 3$, $N = 5$: $\mathrm{Sym}(4)$ has dimension 10, the rank-≤ 3 locus has dimension 9, and after fixing scale the shape space has dimension 8.

2. **There is no determinant.** $\det X$ is undefined, so the orientation coordinate of Theorem 2.4 does not exist and the branched-cover description of the rotation-reduced space must be replaced.

3. **The tangent split fails.** The identity (3) is an identity about $\mathrm{Sym}(d)$; on a rank-constrained variety the tangent space is not all symmetric traceless matrices and the eigenvalue/eigenframe decomposition is not a decomposition of it.

We state this as a boundary of validity rather than a conjecture about what replaces it. The natural next object is the rank-d stratum of $\mathrm{Sym}(N-1)$ with its own quotient geometry, which is a different construction and not treated here. One structural statement nevertheless survives off the ladder and is worth recording.

Proposition 8.1 (Threefold split)**.** *At a rank-r point of the trace-one positive semidefinite forms on $\mathbb{R}^n$,*

$$\dim = \frac{r(r+1)}{2} - 1 + r(n-r) \;=\; \underbrace{(r-1)}_{\textit{eigenvalues}} + \underbrace{\frac{r(r-1)}{2}}_{\textit{frame within the support}} + \underbrace{r(n-r)}_{\textit{support-moving}},$$

an identity verified for all (n, r) up to $n = 8$. For five bodies in three dimensions ($n = 4$, $r = 3$): $8 = 2 + 3 + 3$.

The first two blocks are the ladder's split, restricted to the support of G. The third block — motions of the support plane itself — has no analogue on the ladder, and the action of any operator above on it is not determined by the interior theory. That block is the precise boundary of what this paper establishes.

9 Reconstruction

Given a curve $G(t)$ in Σ_d and an initial lift $X(0)$, the physical motion is recovered by lifting horizontally in $P \to \Sigma_d$ and integrating; the accumulated element of $SO(d)$ is the reconstruction holonomy. Nothing in Sections 5–6 depends on this: the operator $\mathcal{A}$ lives on the base and the connection lives on the total space. We note only that the lift is obstructed exactly on the isotropy strata $r \leq d-2$, where the fibre degenerates — at $d=3$, the collinear configurations.

10 Discussion

The content of this paper is that one coincidence, $N = d+1$, is doing all of the organising work, and that once it is made explicit the three-body and four-body constructions stop being analogous and become the same theorem at $d=2$ and $d=3$. The Gram matrix is the invariant; its trace-one slice is the shape space; its determinant is the orientation; and the rotation-reduced space is the double cover branched where that determinant vanishes. The shape sphere, with its radius $1/2$, falls out of this in three lines.

The spectral result is that the transport operator built on this ladder is non-normal but has strictly real, interlacing spectrum on the entire interior, for every d, with the interlacing following from nothing more than positivity of the weights $g_i t_i$ in a secular equation. Exceptional points exist, but only on the boundary divisor, and they are genuine Jordan degeneracies rather than avoided crossings. That is a sharper statement than one usually gets from a non-normal family, and it is available only because the rank-one structure of Proposition 5.4 makes the characteristic polynomial explicit.

Three things are left open and are stated as such. The tensor T has no dynamical derivation (Section 7). The off-ladder case $N-1 > d$ requires a different construction (Section 8). And the relation between the Frobenius structure used here and the kinetic metric of Section 2.4 has been established only at $d=2$; whether the transport spectrum computed in one is the transport spectrum of the other, for $d \geq 3$, has not been checked.

11 The volume structure

Section 2.4 had to name three competing metrics and legislate between them, and Proposition 5.5 shows the results above needed none of them. This section removes the metric from the primitives entirely and lets it be derived, and the consequence is that the free parameter T stops being free.

11.1 The potential, and its first three derivatives

For a regular convex cone $\mathcal{C}$ with a volume form on the dual, the Koszul–Vinberg characteristic function is $\varphi(x) = \int_{\mathcal{C}^*} e^{-\langle x,y\rangle} \mathrm{d}y$, which needs a volume form and no metric. It is relatively invariant, $\varphi(\gamma x) = |\det\gamma|^{-1}\varphi(x)$ for $\gamma \in \mathrm{Aut}(\mathcal{C})$, so $\mathrm{d}^2 \log\varphi$ is an $\mathrm{Aut}(\mathcal{C})$-invariant positive definite form [11, 12]. For the cone of positive definite symmetric matrices $\varphi(G) \propto (\det G)^{-(d+1)/2}$, and since $\det G =$

$(\det X)^2 = \mathrm{Vol}^2$,

$$\log\varphi = -\tfrac{d+1}{2}\log\det G = -(d+1)\log\mathrm{Vol}.$$

The potential is the log volume of the configuration. Writing $\psi = -\log\det G$, its first three derivatives are, each verified symbolically against finite differences,

$$\begin{aligned} \mathrm{d}\psi(H) &= -\mathrm{tr}(G^{-1}H) && \text{the dual coordinate, a covector,} && (6)\\ \mathrm{d}^2\psi(H,K) &= \mathrm{tr}(G^{-1}HG^{-1}K) && \text{the Koszul–Vinberg metric } \langle H,K\rangle_G, && (7)\\ \mathrm{d}^3\psi(H,K,L) &= -2\mathrm{tr}(G^{-1}HG^{-1}KG^{-1}L) && \text{the cubic form.} && (8) \end{aligned}$$

Remark 11.1 (Pairs and cycles). The cubic form is fully symmetric in its three arguments (permutation spread 3×10^{-16}). In a principal frame the metric (7) couples index *pairs* $\{i,j\}$ with weight $\propto 1/(g_ig_j)$, while the cubic form (8) couples index *cycles* $i\to j\to k\to i$ with weight $\propto 1/(g_ig_jg_k)$. In Hessian geometry the pair (metric, cubic form) is the complete local data, the cubic measuring the failure of the flat connection to be Levi-Civita [12]. It has no counterpart in the Frobenius setting, where there is no potential to differentiate a third time.

11.2 The volume metric fixes the parameter

Theorem 11.2 (The parameter is the dual coordinate)**.** *Let G and T be positive definite and simultaneously diagonal. Then $\mathcal{A}(H) = TH + HT - 2\mathrm{tr}(TH)G$ is self-adjoint for $\langle\cdot,\cdot\rangle_G$ if and only if $T \propto G^{-1}$. The same holds for the congruence family $\mathcal{C}(H) = SHS - \mathrm{tr}(S^2H)G$ with $S^2 \propto G^{-1}$.*

Proof. For the Lyapunov part, $\langle TH+HT,K\rangle_G = \sum_{i,j}(t_i+t_j)H_{ij}K_{ij}/(g_ig_j)$, which is already symmetric in H and K. All the asymmetry therefore lies in the trace-removal term, whose contribution is $-2\mathrm{tr}(TH)\mathrm{tr}(G^{-1}K)$; this is symmetric in H,K precisely when T is proportional to G^{-1}. The congruence case is identical with S^2 in place of T and coefficient 1. □

Measured asymmetries confirm it: for $\mathcal{A}$ they are 84.2 at a random T, 14.6 at $T=G$, and 1.8×10^{-12} at $T=G^{-1}$; for $\mathcal{C}$, 35.0, 13.3 and 3.6×10^{-12}.

Remark 11.3 (The non-normality of Section 6 was measured with the wrong form). At $T=G^{-1}$ the operator has Frobenius asymmetry 2.74 and Koszul–Vinberg asymmetry 3.6×10^{-12}, with real spectrum. So an operator can be non-symmetric in the Frobenius structure and self-adjoint in the volume structure at the same time. The remark in Section 7 that the canonical closure $T=G$ destroys non-normality is correct as stated but is an artefact of which form the question was asked in.

11.3 The two canonical operators

Theorem 11.2 leaves two parameter-free operators, one additive and one multiplicative:

$$\mathcal{A}_0(H) = G^{-1}H + HG^{-1} - 2\mathrm{tr}(G^{-1}H)\,G, \tag{9}$$

$$\mathcal{C}_0(H) = G^{-1/2}HG^{-1/2} - \mathrm{tr}(G^{-1}H)\,G. \tag{10}$$

The Lyapunov map is the generator of congruence, $\frac{\mathrm{d}}{\mathrm{d}s}\big|_0 e^{sT/2}He^{sT/2} = \frac12(TH+HT)$, so (10) is the exponentiated form of (9). Two differences are not cosmetic. First, trace closure forces coefficient 1 in (10) against 2 in (9). Second, the eigenframe spectra are $1/g_i + 1/g_j$ for $\mathcal{A}_0$ and $1/\sqrt{g_ig_j}$ for $\mathcal{C}_0$: arithmetic against geometric.

Remark 11.4 (What is a restatement). Setting $S = e^{T/2}$ makes $s_i s_j = e^{(t_i+t_j)/2}$ exactly, so on the eigenframe block alone the multiplicative form is a relabelling of the additive one and carries no new content. The content is in the trace-removal coefficient and, as Section 11.6 shows, in the boundary rates.

Proposition 11.5 (Spectrum of $\mathcal{C}_0$)**.** *$\mathcal{C}_0$ is block diagonal in the splitting* (3)*. Its eigenframe eigenvalues are $1/\sqrt{g_i g_j}$; its eigenvalue block is $D_p - g\,p^\top$ with $p_i = 1/g_i$, so by Theorem 6.2 its spectrum is real and interlaces the p_i, the secular equation reducing to*

$$\sum_{i=1}^{d} \frac{1}{1/g_i - \lambda} = 1$$

because every weight $g_i p_i$ equals 1. Consequently the entire spectrum lies in $[1/g_{\max}, 1/g_{\min}]$: one interlacing structure covers all $\frac{d(d+1)}{2} - 1$ eigenvalues, which is not the case for $\mathcal{A}_0$.

11.4 The multiplicative operator is the square root of the metric

The choice between $\mathcal{A}_0$ and $\mathcal{C}_0$ is not open at the kinematic level, and the reason is one identity.

Theorem 11.6 (Kinematic selection)**.** *Read the volume metric through the trace pairing: the unique operator $\mathcal{M}$ on $\mathrm{Sym}(d)$ with $\mathrm{tr}\big(H\,\mathcal{M}(K)\big) = \langle H, K\rangle_G$ for all H, K is*

$$\mathcal{M}(K) = G^{-1} K G^{-1}.$$

Its unique positive square root (positive and self-adjoint for the trace pairing) is the congruence $K \mapsto G^{-1/2} K G^{-1/2}$ — exactly the congruence part of $\mathcal{C}_0$.

Proof. $\mathrm{tr}(H\,G^{-1}KG^{-1}) = \langle H, K\rangle_G$ is the definition of the metric. The congruence by $G^{-1/2}$ composes with itself to $\mathcal{M}$ (verified to 1.4×10^{-14}), is symmetric for the trace pairing, and is positive because $\mathrm{tr}(H\,G^{-1/2}HG^{-1/2}) = \|G^{-1/4}HG^{-1/4}\|_F^2 > 0$ for $H \neq 0$. A positive operator has exactly one positive square root. □

Remark 11.7 (What this selects, and what it does not). $\mathcal{C}_0$ is a function of the volume metric alone: its square root, trace-corrected to preserve the slice. $\mathcal{A}_0$ is the logarithmic generator of that congruence, one step removed. So the volume structure singles out $\mathcal{C}_0$ with no further choice — a kinematic selection. Whether a *dynamical* principle selects the same operator is a separate question and remains open; the distinction matters and the two claims should not be merged.

11.5 The cubic form is the connection

Theorem 11.8. *In the flat (affine) coordinates of* $\mathrm{Sym}(d)$*, the Levi-Civita Christoffel symbols of the volume metric are*

$$\Gamma_{ijk} = \tfrac{1}{2}\, C_{ijk}$$

exactly, where $C = \mathrm{d}^3\psi$ is the cubic form (8)*. The geodesic equation is $\ddot{G} = \dot{G}\,G^{-1}\dot{G}$, solved in closed form by*

$$G(t) = G_0^{1/2}\, e^{tV}\, G_0^{1/2}, \qquad V \in \mathrm{Sym}(d).$$

Proof. Since $g = \mathrm{d}^2\psi$, the coordinate derivative of the metric is the third derivative of the potential: $\partial_i g_{jk} = C_{ijk}$, fully symmetric (verified over all index triples to 3.1×10^{-12}). Hence $\Gamma_{ijk} = \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) = \frac{1}{2}C_{ijk}$. For the geodesics, $\dot G = G_0^{1/2} V e^{tV} G_0^{1/2}$ and $\ddot G = G_0^{1/2} V^2 e^{tV} G_0^{1/2}$, while

$$\dot G\, G^{-1} \dot G = G_0^{1/2} V e^{tV}\, e^{-tV}\, V e^{tV} G_0^{1/2} = G_0^{1/2} V^2 e^{tV} G_0^{1/2} = \ddot G$$

because V commutes with e^{tV}. Verified numerically to 5.2×10^{-7} by second differences, with the KV speed $\mathrm{tr}\big((G^{-1}\dot G)^2\big)$ conserved to 1.2×10^{-11}. □

So the cubic form is not optional structure sitting beside the metric: it *is* the Christoffel data of volume transport, and it exists only because the metric has a potential — there is nothing to differentiate a third time in the Frobenius setting.

Remark 11.9 (An inversion that must be owned). $\langle H, K\rangle_G = \mathrm{tr}(G^{-1}HG^{-1}K)$ is the affine-invariant metric on the cone of positive matrices — precisely the third metric Section 2.4 warned against importing. It has now arrived anyway, but by derivation rather than importation: it is the Hessian of the log volume, and every appearance of it above is an appearance of the potential. The warning stands as written — do not let a metric in silently — and is satisfied here by naming its origin. The cost is a genuine tension that the paper owns rather than resolves: at $d = 2$ the interior of Σ_d is *hyperbolic* in the volume structure and *round* in the kinetic one. Both origins are principled — mechanics for the kinetic metric, the volume form for this one — they answer different questions, and every statement in this paper says which structure it is made in.

11.6 The boundary, in the volume structure

This subsection replaces Section 6.1; it does not extend it. Pinning at a finite value, the branch collision and the Jordan block have no analogue here.

Theorem 11.10 (Boundary rates)**.** *Let $g_k = \varepsilon \to 0$ with the remaining g_i bounded away from zero. For $\mathcal{C}_0$:*

(i) exactly one eigenvalue diverges, from the eigenvalue block, with $\lambda_{\max} = 1/\varepsilon - 1 + O(\varepsilon)$;

(ii) exactly $d-1$ eigenvalues diverge from the eigenframe block, at rate $\varepsilon^{-1/2}$, with prefactor $1/\sqrt{g_j}$ for the partner index j;

(iii) the remaining $\frac{(d-1)(d-2)}{2} + (d-2)$ eigenvalues remain finite;

(iv) no Jordan block forms: the divergent eigenvalue escapes rather than colliding.

Sketch and verification. For (i), write $\lambda = p_k - \delta$ in the secular equation; the remaining terms contribute $-(d-1)/\lambda \to 0$, giving $1/\delta \to 1$. Measured at $d = 3$, $\varepsilon = 10^{-6}$: $\lambda_{\max} = 1000000.000$ against $1/g_{\min} - 1 = 1000000.000$. For (ii), $1/\sqrt{g_i g_k}$ with one index at ε. Log–log slopes over $\varepsilon \in [10^{-6}, 10^{-3}]$ give -1.00000 for the top eigenvalue and -0.49993 ($d = 3$), -0.49996 ($d = 4$) for the next, against the predicted -1 and $-\frac{1}{2}$. For (iv), the minimum over 400 near-boundary draws of the normalised eigenvector determinant is 0.87, 0.73, 0.60 at $d = 3, 4, 5$: bounded away from zero. □

Corollary 11.11 (In terms of the volume)**.** *Since $\det G = \prod_i g_i = \mathrm{Vol}^2$ and the other eigenvalues stay bounded, $g_{\min}$ is proportional to Vol^2 near the divisor. The eigenvalue block therefore diverges as Vol^{-2} and the eigenframe block as Vol^{-1}.*

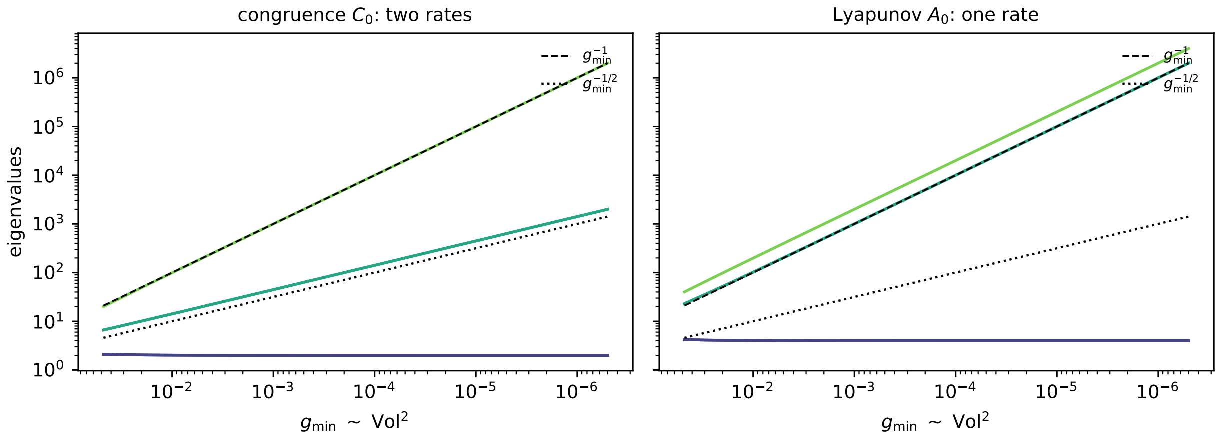

Figure 3: Boundary behaviour of the two canonical operators at $d = 3$, against $g_{\min} \sim \mathrm{Vol}^2$ (axis reversed, degeneracy to the right). The congruence operator $\mathcal{C}_0$ splits into two divergence rates, $g_{\min}^{-1}$ and $g_{\min}^{-1/2}$; the Lyapunov operator $\mathcal{A}_0$ sends everything off at the single rate $g_{\min}^{-1}$. Measured slopes: -1.00000 and -0.49993 for $\mathcal{C}_0$, -1.00000 and -0.99980 for $\mathcal{A}_0$.

Remark 11.12 (The two canonical operators are distinguishable). Theorem 11.10(ii) fails for $\mathcal{A}_0$, whose eigenframe eigenvalues $1/g_i + 1/g_j$ also diverge at rate ε^{-1}; the measured slopes are -1.00000 and -0.99980. So the choice between the additive and the multiplicative canonical operator is not a matter of convention: they have different exponents at the degenerate stratum, and an exponent is measurable.

11.7 The corner of the divisor

Theorem 11.10 treats one vanishing eigenvalue — the smooth part of the divisor. The divisor is not smooth everywhere.

Proposition 11.13 (Corner rates)**.** *The gradient of* $\det G$ *is the adjugate* $\mathrm{adj}(G)$*, which vanishes if and only if* $\operatorname{rank} G \leq d - 2$*. Hence* $\mathcal{D}$ *is smooth exactly on the rank-*$(d-1)$ *stratum and singular exactly on the isotropy stratum. Now let two eigenvalues vanish together,* $g_k = g_l = \varepsilon \to 0$*. For* $\mathcal{C}_0$*:*

(i) the degenerate pair supplies an exact *eigenvalue at* $1/\varepsilon$*, carried by the direction* $e_k - e_l$ *(verified to relative* 1.2×10^{-16}*);*

(ii) the eigenframe pair (k, l) *has eigenvalue* $1/\sqrt{g_k g_l} = 1/\varepsilon$*;*

(iii) the measured log–log slopes of the top three eigenvalues are -0.99973*,* -0.99973*,* -1.00000*:* the two rates of Theorem 11.10 collide into one*.*

Remark 11.14 (The third row). Rate splitting is therefore a property of the smooth part of the divisor: on the codimension-one wall the operator distinguishes Vol^{-2} from Vol^{-1}, and at the corner the distinction disappears. The taxonomy of degenerations running through this programme — the pole (kernel along the locus, no genuine stratum) and the fold (kernel across it, and here two rates) — acquires a third entry: the *corner*, where the divisor itself is singular and the rates merge. Which singularity class the corner belongs to in Arnold's hierarchy is not attempted here.

11.8 What this settles and what it does not

Settled: the parameter T, open since the definition of the operator, is forced to be the dual coordinate G^{-1} once volume is the primitive (Theorem 11.2); the apparent conflict between the canonical closure and non-normality was a wrong-form measurement; the boundary is a divergence in powers of the volume rather than a pinning, with rates that split on the smooth divisor and merge at its corner (Theorem 11.10, Proposition 11.13); the kinematic choice between the canonical operators — $\mathcal{C}_0$ is the square root of the metric and $\mathcal{A}_0$ its logarithmic generator (Theorem 11.6); and the role of the cubic form — it is the connection (Theorem 11.8).

Not settled, and stated as the open boundary of the paper: a *dynamical* derivation — the kinematic selection says which operator the volume structure builds, not which one a mechanical principle would; the support-moving block of Proposition 8.1, on which no operator here acts in a determined way; and the singularity class of the corner.

A Wall vectors at $d = 3$

With Jacobi vectors $\rho_1 = \sqrt{\mu_1}(q_2 - q_1)$, $\rho_2 = \sqrt{\mu_2}(q_3 - c_{12})$, $\rho_3 = \sqrt{\mu_3}(q_4 - c_{123})$ and reduced masses

$$\mu_1 = \frac{m_1 m_2}{m_1 + m_2}, \quad \mu_2 = \frac{(m_1 + m_2)m_3}{m_1 + m_2 + m_3}, \quad \mu_3 = \frac{(m_1 + m_2 + m_3)m_4}{m_1 + m_2 + m_3 + m_4},$$

the vectors $c^{(ij)}$ of (2) are

$$c^{(12)} = \left[\tfrac{1}{\sqrt{\mu_1}},\, 0,\, 0\right], \qquad c^{(23)} = \left[-\tfrac{m_1}{(m_1+m_2)\sqrt{\mu_1}},\, \tfrac{1}{\sqrt{\mu_2}},\, 0\right],$$

$$c^{(13)} = \left[\tfrac{m_2}{(m_1+m_2)\sqrt{\mu_1}},\, \tfrac{1}{\sqrt{\mu_2}},\, 0\right], \qquad c^{(34)} = \left[0,\, -\tfrac{m_1+m_2}{(m_1+m_2+m_3)\sqrt{\mu_2}},\, \tfrac{1}{\sqrt{\mu_3}}\right],$$

$$c^{(24)} = \left[-\tfrac{m_1}{(m_1+m_2)\sqrt{\mu_1}},\, \tfrac{m_3}{(m_1+m_2+m_3)\sqrt{\mu_2}},\, \tfrac{1}{\sqrt{\mu_3}}\right], \qquad c^{(14)} = \left[\tfrac{m_2}{(m_1+m_2)\sqrt{\mu_1}},\, \tfrac{m_3}{(m_1+m_2+m_3)\sqrt{\mu_2}},\, \tfrac{1}{\sqrt{\mu_3}}\right].$$

All six satisfy $\left(c^{(ij)}\right)^{\top} G c^{(ij)} = |q_i - q_j|^2$ identically; checked against a random configuration with unequal masses to 3.6×10^{-15}.

B Verification

The script VERIFY-LADDER-1 accompanies this paper and reproduces every number quoted. All fourteen checks pass.

Second pass

The script VERIFY-LADDER-2 runs the additional checks of this version; 13 of 14 pass and the one failure is recorded here rather than removed, because it corrected a claim.

The failed check was a prediction made in drafting, not a result in the previous version: it was expected that two coinciding pins would also produce a Jordan block. They do not. The eigenvector equation $t{\cdot}v = 0$ then admits a two-dimensional solution space, and Theorem 6.5(iii) is stated accordingly.

Third pass

The script VERIFY-LADDER-3 runs the checks of this version; all ten pass.

Check	Result	
$\dim \Sigma_d = (d-1) + \frac{d(d-1)}{2}$, $d = 2..8$	identity holds	
Collision locus codimension $= d$	identity holds	
$d = 2$: $(u, v, \det X)$ on the sphere of radius $\frac{1}{2}$	max deviation	2.2×10^{-16}
$d = 2$: kinetic metric is round	pushforward Gram $= \mathrm{diag}(1, 1)$	off-diag 6.7×10^{-12}
Six wall vectors reproduce $\lvert q_i - q_j \rvert^2$	max error	3.6×10^{-15}
Trace coefficient $c = 2$ forced	c=1 leaves by 0.693; c=2 by	2.8×10^{-16}
Block diagonality of $\mathcal{A}$	coupling norm	$< 10^{-13}$
Eigenframe eigenvalues $= t_i + t_j$	max error	8.9×10^{-16}
Spectrum real, $d = 2..6$, 15000 draws	$\max \lvert \mathrm{Im}\, \lambda \rvert$	0.0
One secular root per gap, every draw	holds	
Spectrum $= \{t_i + t_j\} \cup \{2\lambda_k\}$	max error	9.2×10^{-14}
Boundary pinning at $2t_k$	holds	
Boundary exceptional point is defective	$\lvert$ det eigvecs$\rvert$	1.1×10^{-15}
Off-ladder dimensions (d=3, N=5)	Sym(4) dim 10, shape dim 8	

Check	Result	
Diagonal similarity $D_t - gt^\top \sim D_t - uu^\top$	max error	1.3×10^{-15}
$T = G \Rightarrow \mathcal{A}$ symmetric	$\lVert \mathcal{A} - \mathcal{A}^\top \rVert$	3.3×10^{-16}
$T \neq G \Rightarrow \mathcal{A}$ non-symmetric	$\lVert \mathcal{A} - \mathcal{A}^\top \rVert$	3.75
Spectrum in a random non-orthonormal basis	max error	2.1×10^{-9}
Pins at $2t_k$ present, 1–3 zeros, $d = 3, 4, 5$	all present	
Generic rank-one boundary not defective	$\lvert$ det eigvecs$\rvert$	0.866
Two coinciding pins	$\lvert$ det eigvecs$\rvert = 1.000$: **semisimple**	(prediction corrected)
EP removed by generic perturbation	$6 \times 10^{-16} \to 3.4 \times 10^{-1}$	

References

[1] D. G. Kendall, Shape manifolds, Procrustean metrics, and complex projective spaces, *Bull. London Math. Soc.* **16**, 81 (1984).

[2] I. L. Dryden, A. Kume, H. Le, A. T. A. Wood, A multidimensional scaling approach to shape analysis, *Biometrika* **95**, 779 (2008). [Reflection shape space embedded in the cone of positive semidefinite matrices.]

[3] R. Montgomery, The three-body problem and the shape sphere, *Amer. Math. Monthly* **122**, 299 (2015).

[4] A. Chenciner and R. Montgomery, A remarkable periodic solution of the three-body problem in the case of equal masses, *Ann. of Math.* **152**, 881 (2000).

[5] A. Albouy and A. Chenciner, Le problème des n corps et les distances mutuelles, *Invent. Math.* **131**, 151 (1998).

[6] R. Moeckel and R. Montgomery, Symmetric regularization, reduction and blow-up of the planar three-body problem, *Pacific J. Math.* **262**, 129 (2013).

[7] T. Kato, *Perturbation Theory for Linear Operators*, Springer (1966).

[8] G. H. Golub and C. F. Van Loan, *Matrix Computations*, Johns Hopkins (2013). [Secular equations and rank-one eigenvalue updates.]

Check	Result	
$\mathcal{C}_{G^{-1/2}} \circ \mathcal{C}_{G^{-1/2}} = \mathcal{M}$ (metric operator)	max error	1.4×10^{-14}
Congruence trace-symmetric and positive	holds	
$\partial_i g_{jk} = C_{ijk}$, all index triples	max error	3.1×10^{-12}
Geodesic closed form $\ddot{G} = \dot{G}G^{-1}\dot{G}$	max error	5.2×10^{-7}
KV speed conserved along geodesics	relative spread	1.2×10^{-11}
$\mathrm{adj}(G) \to 0$ iff rank $\leq d-2$	1.4×10^{-9} vs 0.25	
Corner slopes (two vanishing eigenvalues)	-0.99973, -0.99973, -1.00000	
Exact eigenvalue at $1/g_{\min}$ from the degenerate pair	relative error	1.2×10^{-16}
Threefold dimension identity, all (n, r), $n \leq 8$	holds	
Rank-3 point of Sym(4): three blocks well-defined	holds	

[9] G. H. Golub, Some modified matrix eigenvalue problems, *SIAM Review* **15**, 318 (1973).

[10] J. R. Bunch, C. P. Nielsen and D. C. Sorensen, Rank-one modification of the symmetric eigenproblem, *Numer. Math.* **31**, 31 (1978).

[11] E. B. Vinberg, The theory of convex homogeneous cones, *Trans. Moscow Math. Soc.* **12**, 340 (1963).

[12] H. Shima, *The Geometry of Hessian Structures*, World Scientific (2007).

[13] J.-L. Koszul, Domaines bornés homogènes et orbites de groupes de transformations affines, *Bull. Soc. Math. France* **89**, 515 (1961).

[14] R. Bhatia, *Positive Definite Matrices*, Princeton University Press (2007).